Artificial Intelligence & Machine Learning – AIML

(Empowering the Next Generation of AI Engineers)

AI/ML Training (AIML) program transforms learners into job-ready AI/ML engineers. Through hands-on projects and real-world applications, it takes you from core fundamentals to advanced AI/ML skills that employers value.

Industry-Ready AI Skills:

🤖 **Python for AI/ML:** Master AI-focused Python programming to build real-time, production-ready AI/ML applications.
🤖 **Machine Learning (ML):** Learn end-to-end ML workflows including data preparation, feature engineering, model training, evaluation, and inference.
📦 **Deep Learning:** Build and train neural networks to solve complex problems using modern deep learning techniques.
☁️ **Generative AI (GenAI):** Get hands-on experience with **Transformers and Large Language Models (LLMs)** to build practical, real-world GenAI solutions.
💡 **Prompt Engineering:** Design high-quality prompts to improve LLM accuracy, efficiency, and real-world usability.
🧠 **Agentic AI:** Create intelligent AI agents and real-time workflows using modern agentic AI frameworks.

Technology Focus:

Python Foundations : Build a strong programming base using **NumPy, Pandas, REST APIs, and Object-Oriented Programming (OOPs)** for AI/ML development.
🤖 **Machine Learning:** Implement real-world ML solutions using **Scikit-learn**, covering core algorithms for classification, regression, and model evaluation.
🧠 **Deep Learning:** Design and train neural networks using **PyTorch and TensorFlow** for computer vision and deep learning applications.
🗣️ **NLP, Generative AI & Agentic AI:** Develop intelligent AI systems using **GPT, BERT, LLaMA, and Claude LLMs**, along with **RAG architectures, LangChain**, and agent-based workflows.
🚀 **Hands-On Projects:** Gain practical experience through end-to-end projects including **Classification, Regression, Text Generation, Image Detection**, and real-world AI **ChatBot** use cases.

Highlights:

Career Support: Resume reviews, mock interviews, and guidance to land your first AI/ML role.
Comprehensive Curriculum: From Python basics to deep learning, NLP, and recommendation systems.
Latest Technology: Explore Generative AI, ChatGPT integration, and reinforcement learning.
Industry-Aligned: Designed with inputs from AI/ML professionals and top tech companies.
Technical Support: Tech-talk sessions to clarify doubts and get extra help.
Interactive Learning: Live coding and mini-assignments after each module for hands-on practice.
Flexible Pace: Learn at your own speed with lifetime access to videos.
Real-World Projects: Work on industry-relevant datasets and projects.
Job-Ready Skills: Master Python, Machine Learning, Agentic AI, Gen-AI, and AI/ML Operations.

Module - 1: PYTHON

Chapter 1: Python Fundamentals

Python installation, environment setup (venv, pip)
Python Overview, Features, Environment Setup, IDEs (PyCharm), Jupyter Notebook

Code Execution, Syntax, Comments, Indentation
Data Types (int, float, str, bool), Strings, Type Conversion
Operators - Arithmetic, Relational, Logical, Assignment, Membership, Identity, String Concatenation

Chapter 2: Flow Control & Collections

Conditional Statements (if, if-else, if-elif, nested if), Looping (while, for, break, continue, pass)
Data Structures - Lists, Tuples, Dictionaries, Sets, zip(), comprehensions
Methods on List, Tuple, Dictionary, Set, Practice exercises, enumerate

Chapter 3: Built-ins, Functions & File Handling

Built-in Functions - range(), len(), id(), type(), dir(), str(), etc.
User-Defined Functions - Defining, calling, args, kwargs, return, default values, Call by value/reference
File Operations – Reading, Writing, Modes - csv, xml, json, xlsx
Functional Programming - lambda, map(), filter()
Regular Expressions - Pattern Matching, Search/Substitute/Split

Chapter 4: Testing, Debugging & Exceptions

Enhance Logging in Python Development
Learn how to program using logging module
Common debugging techniques and tools.
Writing and running unit tests with unittest and pytest.
Exception Handling - try-except, finally, custom exceptions
Testing Tools: Postman & Swagger

Chapter 5: Modules & Packages

Modules & Namespaces - import, __init__.py, project structure, locals(), globals()
Understand how you can write modular code building blocks to reuse functional units of code.

Chapter 6: Python OOPs

OOP Concepts - Class, Object, Constructor (__init__), Encapsulation, Inheritance, Polymorphism, Abstraction, Method overloading/overriding, Class methods

Chapter 7: Database Integration

MySQL: CURD operations (Create, Update, Read, Delete)
FastAPI integration with MySQL

Chapter 8: Session Management & async programming

Session Management: Handling user sessions, tokens, and context flow.
Async Programming: Using async/await for concurrent tasks.
High-Performance Agents: Fast, non-blocking tool and API execution.
JWT Token Management

Chapter 9: Scripting Modules

System Admin Modules with Python (os, subprocess, glob, sys) - File systems, shell scripts

Chapter 10: API Development

FastAPI fundamentals and asynchronous web applications.
Creating and consuming RESTful APIs.
Structured outputs, JSON schemas
Validating outputs with pydantic
Bearer Token authentication for APIs

Module - 2: MACHINE LEARNING

Chapter 1: Machine Learning Fundamentals

What is Machine Learning
Types of Machine Learning: Supervised, Unsupervised, Semi-Supervised
ML vs AI vs LLM vs Data Science vs MLOps
ML Terminologies: Features, Labels, Training, Testing
Role of Data Engineering in ML pipelines

Chapter 2: Data Preparation & Feature Engineering

Data Collection & Cleaning
Handling Missing Values
Feature Scaling & Encoding
Train/Test Split
Data Validation Concepts

Chapter 3: Machine Learning Algorithms (Scikit-learn)

Linear Regression & Polynomial Regression
Logistic Regression
Decision Trees & Random Forest
Model Evaluation Metrics (Accuracy, Precision, Recall, F1-Score)

Chapter 4: Model Training & Inference

Model Training Workflow
Hyperparameter Tuning
Model Inference & Predictions
Model Serialization & Artifacts

Module - 3: AGENTIC AI

Chapter 1: Agentic AI Fundamentals

Introduction to Generative AI, Prompt Engineering & AI Tools
What is Agentic AI: autonomous AI agents and their applications
Key characteristics of agentic systems
Overview of AI tools and frameworks for agentic workflows

Chapter 2: Agentic AI Development

Frameworks overview: LangChain, LangGraph, MCP (Model Context Protocol)
Multi-step reasoning, Orchestrator design
Tools, resources and memory registration with MCP
Designing chatbots that can reason across multiple steps
Integrating external tools (APIs, databases) to perform tasks
Maintaining context and memory across conversations using MCP

Module - 4: GEN-AI

Chapter 1: Generative AI Fundamentals

What is Generative AI
Gen-AI Architecture Overview
Role of LLMs in Gen-AI
Hosted vs Open-Source LLMs

Chapter 2: Large Language Models (LLMs)

LLM Fundamentals & Transformers
GPT, BERT, Claude, LLaMA Overview
Hugging Face Transformers
LLM Inference & Optimization
Working with External LLMs – Integrating Groq LLaMA Models for High-Speed Reasoning
Building Production Agents – Using AWS Bedrock Claude Models for Enterprise LLM Capabilities

Chapter 3: Prompt Engineering

What is Prompt Engineering
Prompt Types: Zero-shot, One-shot, Few-shot
Chain-of-Thought Prompting
Prompt Optimization Techniques

Chapter 4: Retrieval-Augmented Generation (RAG)

What is RAG and Why it Matters
Embeddings & Vector Databases
ChromaDB Architecture
RAG Workflow with LangChain
LLM + RAG Integration

Module-5: AI/ML OPERATIONS

Chapter 1: AI/ML Lifecycle & MLOps

End-to-End AI/ML Lifecycle
Model Planning, Training & Versioning
ML vs DevOps vs MLOps
Model Governance Basics

Chapter 2: Model Packaging & Versioning

Scikit-learn Model Packaging
Deep Learning Model Artifacts
Model Serialization Techniques
Data & Model Versioning using Git & DVC

Chapter 3: Experiment Tracking & Pipelines

MLflow for Experiment Tracking
Model Registry Concepts
Kubeflow Architecture & Pipelines
SageMaker Pipelines & Experiments

Chapter 4: Model Deployment & Serving

Deploy ML Models as REST APIs
FastAPI for Model Serving
Streamlit for Model UI
Real-Time vs Batch Inference

Chapter 5: Cloud & Gen-AI Platforms

Amazon SageMaker Overview
Amazon Bedrock for LLM Hosting
LLM Integration with Bedrock
Secure & Scalable AI Deployments

Module-6: PROJECTS

Chapter 1: Capstone Project - ValAgent – Full Agentic AI System

MCP based Multi-Reasoning Agent development integrated into the streamlit chatbot.
Build a multi-agent AI assistant for Valaxy using orchestrator planning, intent detection, and tool-based execution across microservices.
Integrate Groq/Claude LLMs, RAG search, and ChromaDB to handle courses, enrollment, payments, discounts, and notifications end-to-end.
Deploy with Docker, secure APIs, and real-world monitoring.

Module-7: CAREER GUIDANCE

Chapter 1: Career Optimization

Career Optimization & Guidance: Resume building, LinkedIn polishing, and personalized career roadmap.
Mock Interviews: Weekly technical + HR interview practice with feedback.
Hands-on Support: Weekly assignments, project reviews, and ongoing development mentorship.